

Lecture 3: Pixels and Filters

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What we will learn today

- Images as functions
- Linear systems (filters)
- Convolution and correlation

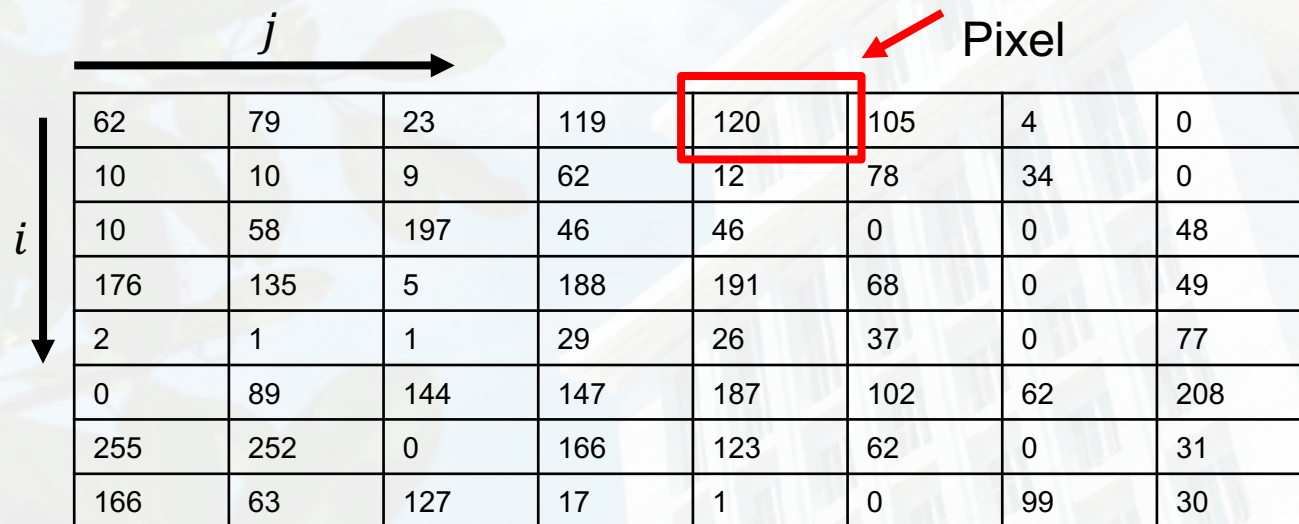


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Images as functions

- Images are usually **digital (discrete)**:
 - **Sample** the 2D space on a regular grid
- Represented as a matrix of integer values



62	79	23	119	120	105	4	0
10	10	9	62	12	78	34	0
10	58	197	46	46	0	0	48
176	135	5	188	191	68	0	49
2	1	1	29	26	37	0	77
0	89	144	147	187	102	62	208
255	252	0	166	123	62	0	31
166	63	127	17	1	0	99	30



Images as functions

Cartesian coordinates

$f[n, m]$ =

↑
Notation for discrete functions

$$\begin{bmatrix} \ddots & & \vdots & & \\ \dots & f[-1, -1] & f[-1, 0] & f[-1, 1] & \\ & f[0, -1] & \underline{f[0, 0]} & f[0, 1] & \dots \\ & f[1, -1] & f[1, 0] & f[1, 1] & \\ & & \vdots & & \ddots \end{bmatrix}$$



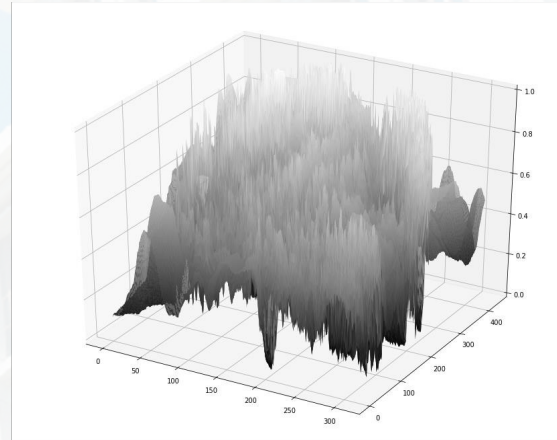

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Images as functions

- An image as a function f from $\mathbb{R}^2$ to $\mathbb{R}^N$
 - $f(x, y)$ gives the **intensity** at position (x, y)
 - Defined over a rectangle, with a finite range:

$$f: \underbrace{[a, b] \times [c, d]}_{\text{Domain Support}} \rightarrow \underbrace{[0, 255]}_{\text{Range}}$$



Images as functions

- An image as a function f from $\mathbb{R}^2$ to $\mathbb{R}^N$
 - $f(x, y)$ gives the **intensity** at position (x, y)
 - Defined over a rectangle, with a finite range:

$$f: \underbrace{[a, b] \times [c, d]}_{\text{Domain Support}} \rightarrow \underbrace{[0, 255]}_{\text{Range}}$$

- A color image: $f(x, y) = \begin{bmatrix} r(x, y) \\ g(x, y) \\ b(x, y) \end{bmatrix}$



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Applications of Linear Systems and Filters

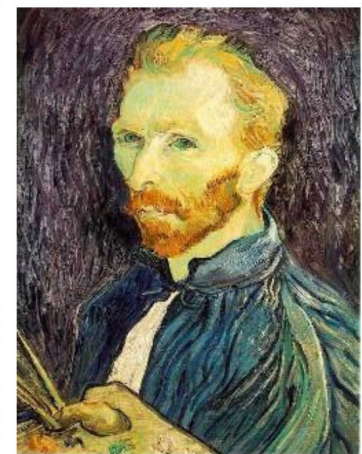
De-noising



Salt and pepper noise



Super-resolution



In-painting



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Systems and Filters

Filtering:

Form a new image whose pixel values are transformed from original pixel values

Goals:

Extract useful information from images

- Features (edges, corners, blobs...)

Modify or enhance image properties

- Super-resolution; in-painting; de-noising



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Systems and Filters

$\mathcal{S}$ is the system operator, defined as a mapping/assignment that transforms the input f into the output g .

$$f[n, m] \rightarrow \boxed{\text{System } \mathcal{S}} \rightarrow g[n, m]$$

$$g = \mathcal{S}[f], \quad g[n, m] = \mathcal{S}\{f[n, m]\}$$

$$f[n, m] \xrightarrow{\mathcal{S}} g[n, m]$$



Filter example #1: Moving Average

2D moving average over a 3×3 neighborhood window

Original image



Smoothed image



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Filter example #1: Moving Average

2D moving average over a 3×3 neighborhood window

$$g[n, m] = \frac{1}{9} \sum_{k=n-1}^{n+1} \sum_{l=m-1}^{m+1} f[k, l]$$

Sum over rows (blue arrow) and Sum over columns (red arrow) are indicated for the summation terms.

$$= \frac{1}{9} \sum_{k=-1}^1 \sum_{l=-1}^1 f[n-k, m-l]$$

Filter "kernel", "mask"

$$\frac{1}{9} \mathbf{h}$$

1	1	1
1	1	1
1	1	1



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Filter example #1: Moving Average

In summary:

- This filter “transforms” each pixel value into the average value of its neighborhood
- Achieve smoothing effect (remove sharp)



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Filter example #2: Image Segmentation

Image segmentation based on a simple threshold:

$$g[n, m] = \begin{cases} 1, & f[n, m] > 100 \\ 0, & \text{otherwise.} \end{cases}$$



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Systems can be classified based on their properties

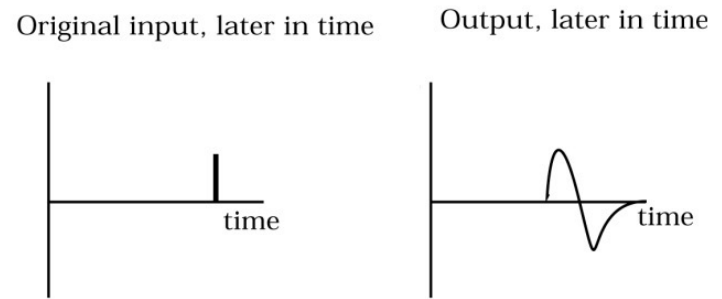
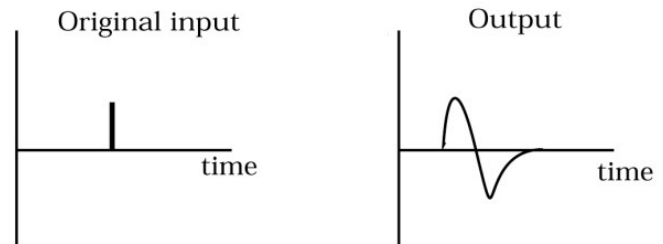
Amplitude properties

- **Linearity**
- Stability
- Invertibility

Spatial properties

- Causality
- Separability
- Memory
- **Shift invariance**
- Rotation invariance

Shift invariance



Linear Shift Invariant (LSI) systems

Linearity

$$S\{\alpha f_1[n, m] + \beta f_2[n, m]\} = \alpha S\{f_1[n, m]\} + \beta S\{f_2[n, m]\}$$

Shift invariance

$$f[n - n_0, m - m_0] \xrightarrow{S} g[n - n_0, m - m_0]$$

For a system to be shift-invariant (or time-invariant) means that a time-shifted version of the input yields a time-shifted version of the output:

$$y(t) = T[x(t)]$$

$$y(t - s) = T[x(t - s)]$$

The response $y(t - s)$ is identical to the response $y(t)$, except that it is shifted in time.



Linear Shift Invariant (LSI) systems

- An LSI system is completely specified by its impulse response h .

$$\delta_2[n, m] \rightarrow \boxed{\mathcal{S}} \rightarrow h[n, m]$$

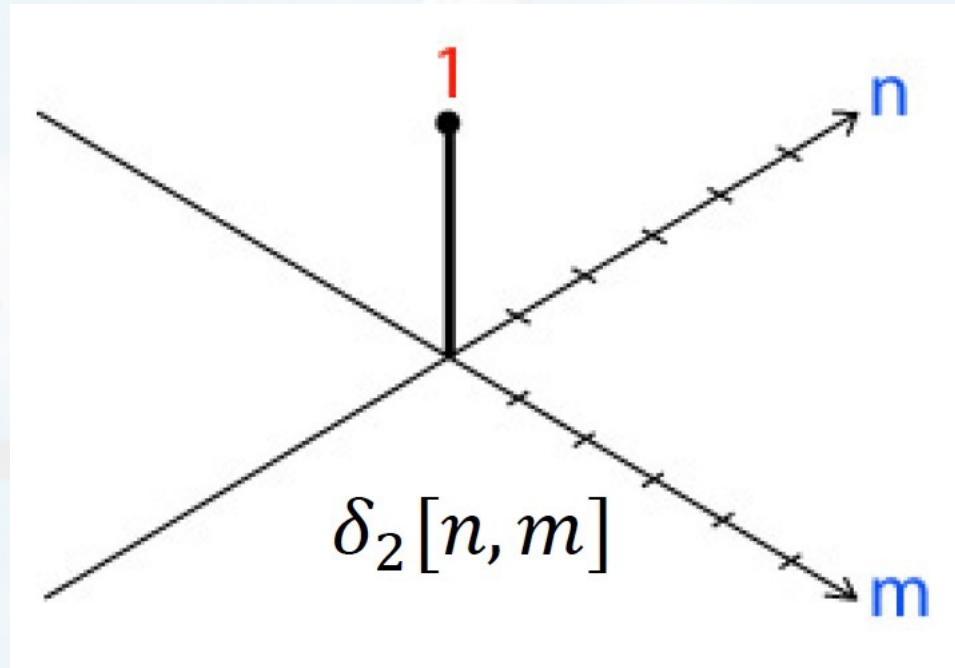
$$\delta_2[n - k, m - l] \rightarrow \boxed{\mathcal{S} \text{ (SI)}} \rightarrow h[n - k, m - l]$$

- By passing an impulse function into an LSI system, we get its impulse response.



2D Impulse Function

- 1 at [0,0]
- 0 everywhere else



Impulse response to the moving average filter

- Recall the expression for our 3x3 moving average filter:

$$f[n, m] \xrightarrow{S} \frac{1}{9} \sum_{k=-1}^1 \sum_{l=-1}^1 f[n-k, m-l]$$

- We can use it to obtain an expression for the impulse response

$$\begin{aligned} \delta_2[n, m] &\xrightarrow{S} h[n, m] \\ &= \frac{1}{9} \sum_{k=-1}^1 \sum_{l=-1}^1 \delta_2[n-k, m-l] \end{aligned}$$



Impulse response of the 3 x 3 moving average filter

$$h[n, m] = \frac{1}{9} \sum_{k=-1}^1 \sum_{l=-1}^1 \delta_2[n - k, m - l]$$

$$= \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix}$$

$\frac{1}{9}$ h

1	1	1
1	1	1
1	1	1



Linear Shift Invariant (LSI) systems

- An LSI system is completely specified by its impulse response h .
 - For any input f , we can compute the output g in terms of the impulse response h .

$$f[n, m] \xrightarrow{S} g[n, m]$$

- In the following, we'll derive an expression for g in terms of h .
- *We know the LSI system satisfies the linearity property and the shift invariance property. We also know h :*

$$\delta_2[n, m] \xrightarrow{S} h[n, m]$$



Key idea: write down f as a sum of impulses

Let's say our input is a 3 x 3 image:

$f[0,0]$	$f[0,1]$	$f[1,1]$
$f[1,0]$	$f[1,1]$	$f[1,2]$
$f[2,0]$	$f[2,1]$	$f[2,2]$

$$=$$

$f[0,0]$	0	0
0	0	0
0	0	0

$$+$$

0	$f[0,1]$	0
0	0	0
0	0	0

$$+ \dots +$$

0	0	0
0	0	0
0	0	$f[2,2]$

$$=$$
 $f[0,0]^*$

1	0	0
0	0	0
0	0	0

$$+$$
 $f[0,1]^*$

0	1	0
0	0	0
0	0	0

$$+ \dots + f[2,2]^*$$

0	0	0
0	0	0
0	0	1

$$= f[0,0] \cdot \delta_2[n,m] + f[0,1] \cdot \delta_2[n,m-1] + \dots + f[2,2] \cdot \delta[n-2,n-2]$$



Key idea: write down f as a sum of impulses

- More generally:

$$f[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot \delta_2[n - k, m - l]$$

- We can now use linearity to see what the output is:

$$f[n, m] \xrightarrow{S} g[n, m]$$

$$\begin{aligned} f[n, m] &= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot \delta_2[n - k, m - l] \\ &\xrightarrow{S} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot S\{\delta_2[n - k, m - l]\} \end{aligned}$$



Key idea: write down f as a sum of impulses

- We have

$$\begin{aligned} f[n, m] &= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot \delta_2[n - k, m - l] \\ &\xrightarrow{S} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot S\{\delta_2[n - k, m - l]\} \end{aligned}$$

- Recall, by the shift invariance property that:

$$S\{\delta_2[n - k, m - l]\} = h[n - k, m - l]$$



Key idea: write down f as a sum of impulses

- We have

$$\begin{aligned} f[n, m] &= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot \delta_2[n - k, m - l] \\ &\xrightarrow{S} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot S\{\delta_2[n - k, m - l]\} \end{aligned}$$

- Which means,

$$f[n, m] \xrightarrow{S} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n - k, m - l]$$



Linear Shift Invariant (LSI) systems

- An LSI system is completely specified by its impulse response h .
 - For any input f , we can compute the output g in terms of the impulse response h .

$$f[n, m] \xrightarrow{S} g[n, m]$$

$$f[n, m] \xrightarrow{S} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n - k, m - l]$$

Discrete convolution

$$f[n, m] * h[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n - k, m - l]$$



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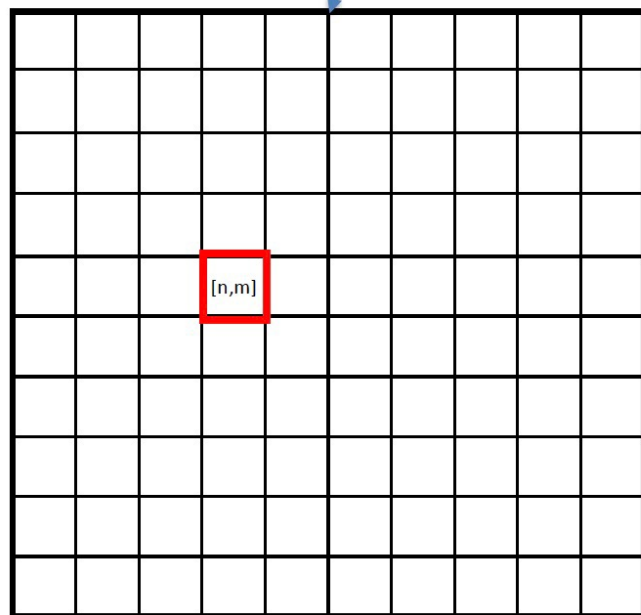


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2D Discrete Convolution

$$f[n, m] * h[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n - k, m - l]$$



Output $f * h$

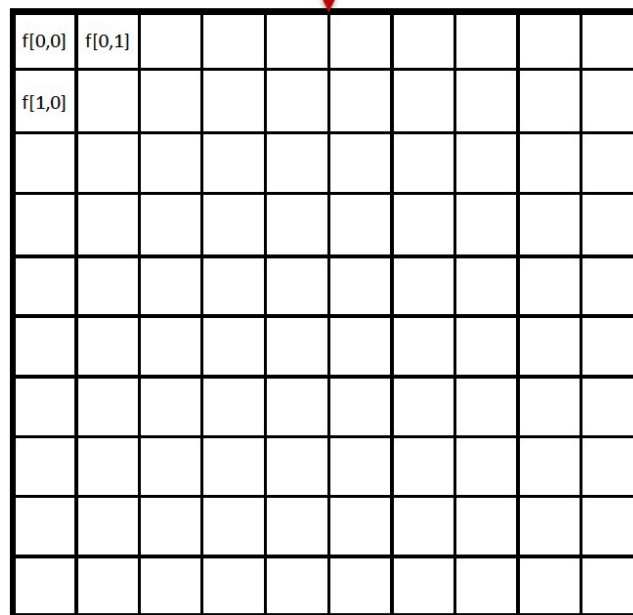


Image $f[k, l]$

$h[-1,-1]$	$h[-1,0]$	$h[-1,1]$
$h[0,-1]$	$h[0,0]$	$h[0,1]$
$h[1,-1]$	$h[1,0]$	$h[1,1]$

Kernel $h[k, l]$

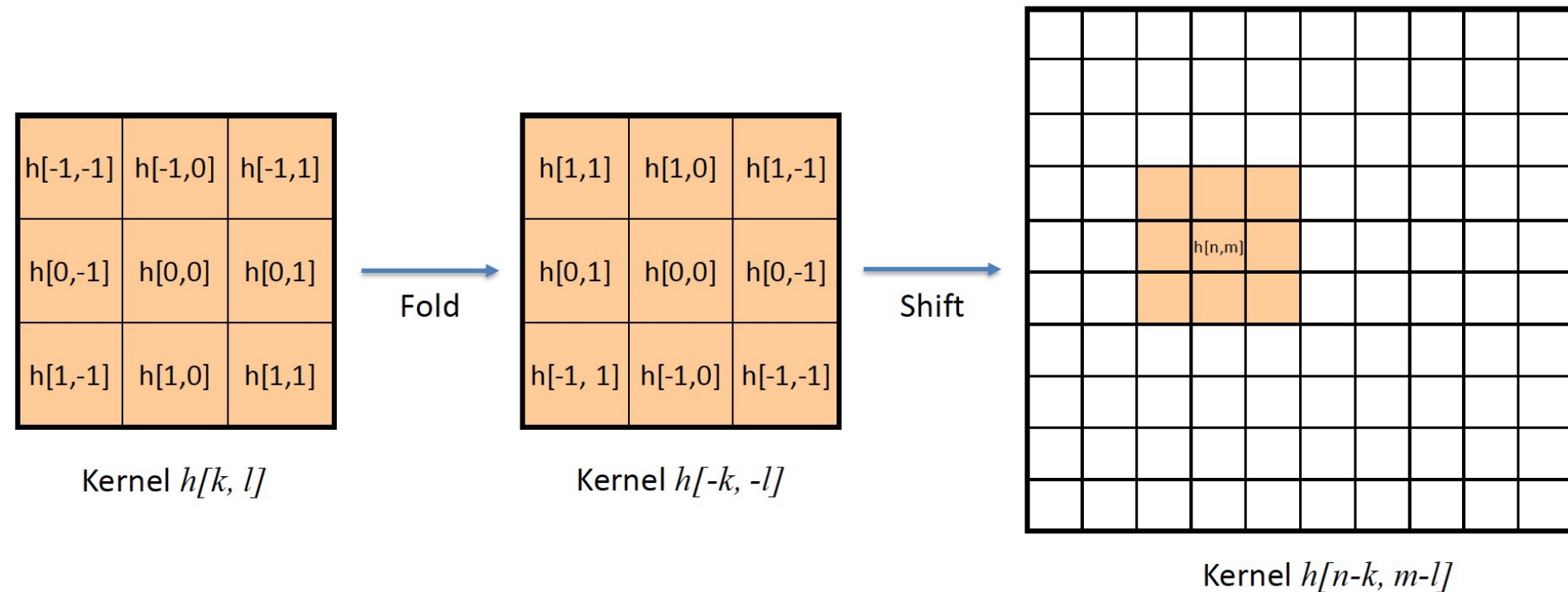


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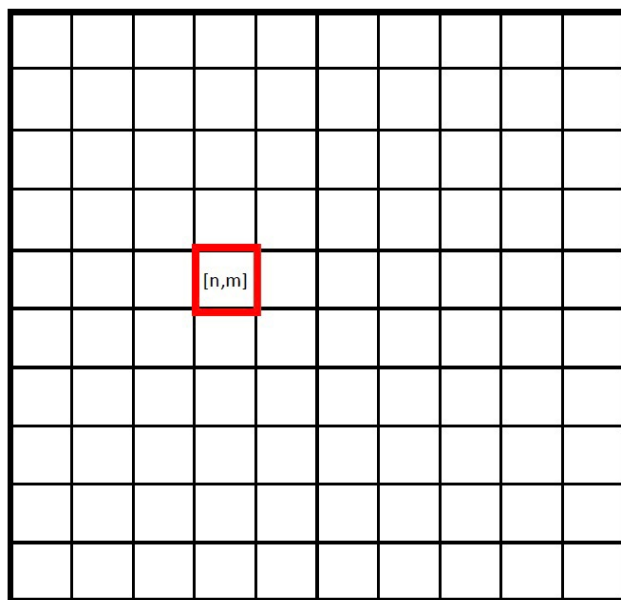
2D Discrete Convolution

$$f[n, m] * h[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n - k, m - l]$$



2D Discrete Convolution

$$f[n, m] * h[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n - k, m - l]$$



Output $f * h$

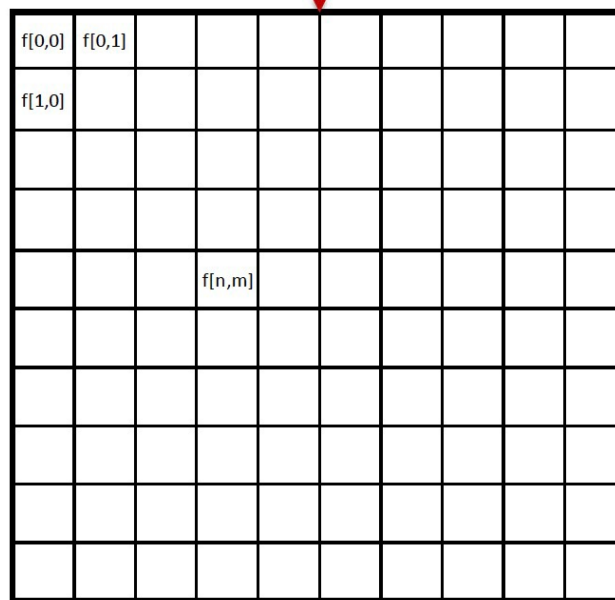
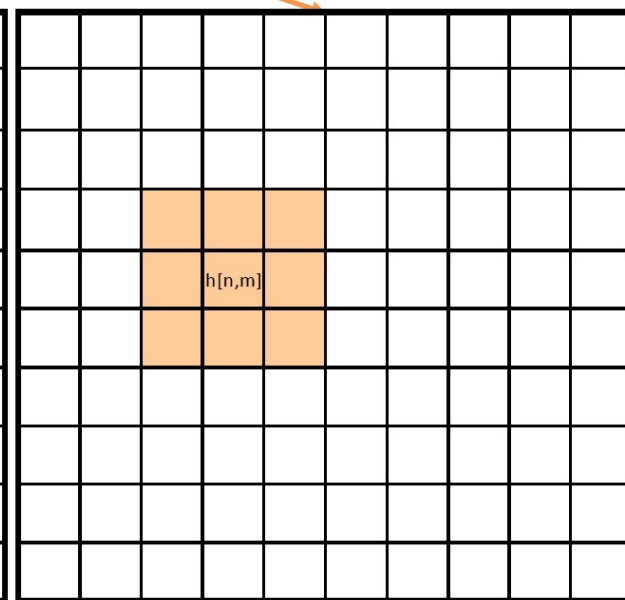


Image $f[k, l]$



Kernel $h[n-k, m-l]$



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2D Discrete Convolution

$$f[n, m] * h[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n - k, m - l]$$

Algorithm:

- Fold $h[k, l]$ about origin to form $h[-k, -l]$
- Shift the folded results by n, m to form $h[n - k, m - l]$
- Multiply $h[n - k, m - l]$ by $f[k, l]$
- Sum over all k, l
- Repeat for every n, m



Convolution in 2D -- examples



Original

*

•0	•0	•0
•0	•1	•0
•0	•0	•0

=

?



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Convolution in 2D -- examples



Original



•0	•0	•0
•0	•1	•0
•0	•0	•0



Filtered
(no change)



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Convolution in 2D -- examples



Original

*

•0	•0	•0
•0	•0	•1
•0	•0	•0

=

?



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Convolution in 2D -- examples



Original



•0	•0	•0
•0	•0	•1
•0	•0	•0



Shifted right
By 1 pixel



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Convolution in 2D -- examples



Original

$$* \left(\begin{bmatrix} \bullet 0 & \bullet 0 & \bullet 0 \\ \bullet 0 & \bullet 2 & \bullet 0 \\ \bullet 0 & \bullet 0 & \bullet 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \end{bmatrix} \right) = ?$$

(Note that filter sums to 1)



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Convolution in 2D -- examples



Original

$$* \left(\begin{bmatrix} \bullet 0 & \bullet 0 & \bullet 0 \\ \bullet 0 & \bullet 2 & \bullet 0 \\ \bullet 0 & \bullet 0 & \bullet 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \end{bmatrix} \right) = ?$$

(Note that filter sums to 1)

“details of the image”

$$\begin{bmatrix} \bullet 0 & \bullet 0 & \bullet 0 \\ \bullet 0 & \bullet 1 & \bullet 0 \\ \bullet 0 & \bullet 0 & \bullet 0 \end{bmatrix} + \begin{bmatrix} \bullet 0 & \bullet 0 & \bullet 0 \\ \bullet 0 & \bullet 1 & \bullet 0 \\ \bullet 0 & \bullet 0 & \bullet 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \end{bmatrix}$$



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What does blurring take away?



- Let's add it back:



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(Cross) Correlation – symbol: **

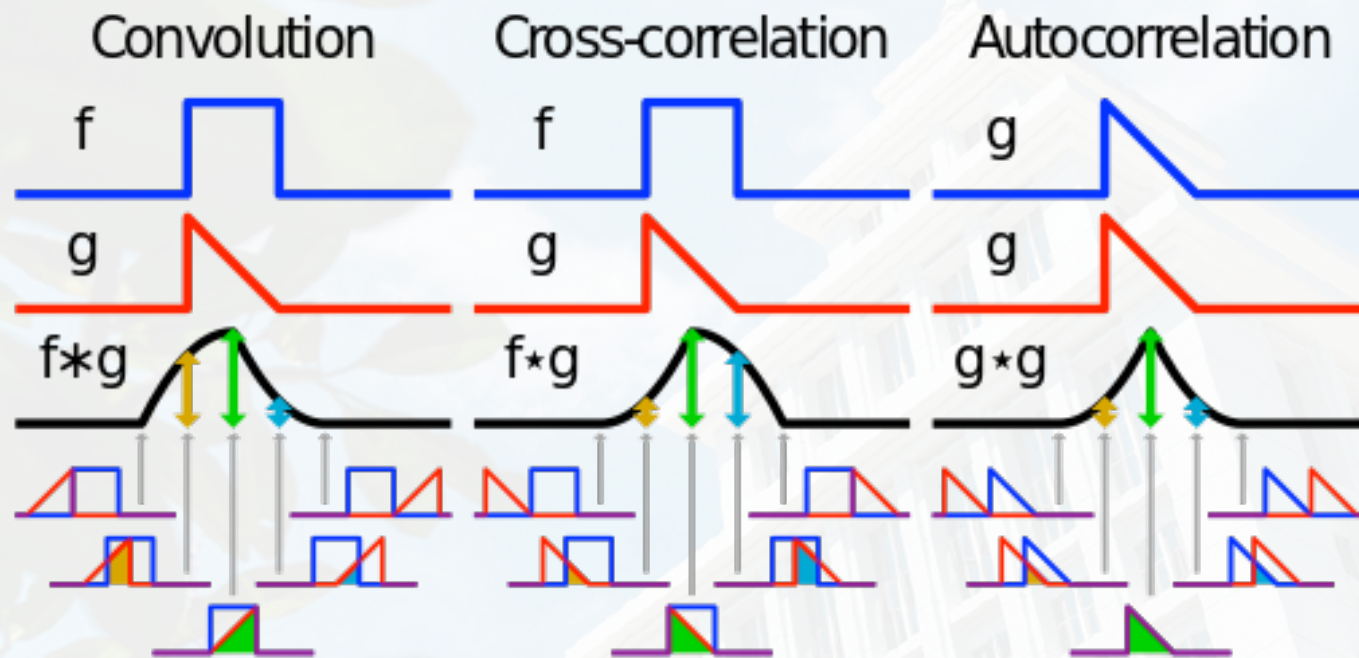
Cross correlation of two 2D signals $f[n,m]$ and $h[n,m]$

$$f[n, m] ** h[n, m] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} f[k, l] \cdot h[n + k, m + l]$$

- Equivalent to a convolution without the flip
- Use it to measure ‘similarity’ between f and h



Convolution vs (Cross) Correlation



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Convolution vs (Cross) Correlation

- **Convolution** is an integral that expresses the amount of overlap of one function as it is shifted over another function.
 - convolution is a filtering operation
- **Correlation** compares the similarity of two sets of data. Correlation computes a measure of similarity of two input signals as they are shifted by one another. The correlation result reaches a maximum at the time when the two signals match best.
 - correlation is a measure of relatedness of two signals



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